

The State at the Origin Is a Matrix

Indeterminate 0/0 forms as physical state functions: point, line, matrix

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L.O.R.E. Framework (Law of Repulsive Emanation)

Abstract

A 0/0 (or 0/0-type indeterminate) form at a coincidence point is not a hole in the state but a STATE whose content is its differential structure. As the position x runs over all reals and the observable is evaluated at a common zero, the ratio $f = g/h$ is undefined as a scalar (the POINT); approached along one direction it becomes the directional slope g'/h' (the LINE); collecting every direction it becomes the linear-response operator - the Jacobian, Onsager flux-force matrix, or Hessian (the MATRIX). Measured on the demon trap at the control origin (where $\Delta\mu = -\ln J = 0$), the 0/0 of the two-ledger ratio is a genuine 2x2 linear-response matrix whose determinant ~ 0 : it is RANK-1, so the two ledgers respond as ONE quantity in both input directions - the "two ledgers one price" of Ch.85 literalised as a matrix degeneracy. The state at the origin is a tensor, not a hole.

1. Point, Line, Matrix

The derivative is the canonical 0/0: $f'(x) = [f(x+h)-f(x)]/h$ at $h = 0$ is 0/0, and its removable value is a slope - a LINE through the origin. In several variables the same ratio becomes a linear map: each approach direction gives one directional derivative, and the full set is a MATRIX (the Jacobian); the Hessian when the observable is a gradient; the Onsager flux-force matrix at equilibrium, where currents and forces both vanish). So:

POINT : $f = g/h$ at the common zero - undefined scalar, $x = 0$

LINE : f along one direction $\rightarrow g'(0)/h'(0)$ - the slope

MATRIX: $R_{ij} = d(\text{output}_i)/d(\text{input}_j)$ - the linear response

This is precisely how physics turns a removable singularity into a state: a transport coefficient, a susceptibility, a conductance are each the well-defined value of a 0/0 ratio at equilibrium.

2. The Demon Strand's 0/0 Origin

At the control (origin) the feedback has no effect: $\Delta\mu = \mu - \mu_0 = 0$ and $-\ln J = 0$ together, so their ratio is 0/0. Build the state's 2x2 linear-response matrix from two outputs ($\Delta\mu$, $-\ln J$) against two genuine input directions (coin coupling c , protocol speed v), each by finite differences:

$$R = \begin{bmatrix} d(\mu)/dc & d(\mu)/dv \\ d(-\ln J)/dc & d(-\ln J)/dv \end{bmatrix}$$

3. Measured Matrix

$$R = \begin{bmatrix} -0.0471 & +0.0355 \\ -0.0515 & +0.0314 \end{bmatrix}$$

$$\det = +0.0004 \quad ; \quad \text{coin-column ratio } d(\mu)/d(-\ln J) = +0.91$$

$$\text{speed-column ratio} = +1.13$$

The coin column ratio ~ -1 recovers the price of Ch.85 along the approach LINE. The determinant ~ 0 makes R RANK-1: the two output columns are nearly collinear, so $\Delta\mu$ and $-\ln J$ move together in BOTH directions - the two ledgers are literally one quantity at the origin. The OFF-diagonal entries are finite, which is what makes it a genuine 2x2 tensor rather than a bare number.

Main Results

THEOREM

The Origin Is a Matrix: a 0/0 at a coincidence point defines a state whose full content is its linear-response matrix (point $\rightarrow$ line $\rightarrow$ matrix); measured at the control origin the two-ledger 0/0 is a genuine 2x2 response tensor with finite cross-entries.

COROLLARY

Rank-1 = One Price: $\det R \sim 0$, so the two outputs respond collinearly in every direction - the "two ledgers one price" of Ch.85 is a MATRIX DEGENERACY, and the coin-column ratio ~ -1 is the price along the approach line.

COROLLARY

Undefined Is Determined: an indeterminate form is not a breakdown; its removable value is a slope (line) and its full structure is a tensor (matrix), exactly as a transport coefficient is the well-defined value of a 0/0 ratio at equilibrium.

Conclusion

The framework's founding 0/0 - the removable singularity as a Law - is here given its proper mathematical-physics content: at the point where a ratio is indeterminate, the state is not lost but revealed as its differential structure, the derivative-slope line and the linear-response matrix. Measured on the trap, the two-ledger 0/0 is a rank-1 tensor, the "one price" made concrete as a degeneracy.

References

- [1] Onsager, L. (1931). Phys. Rev. 37, 405 (flux-force reciprocal relations).
- [2] Kubo, R. (1957). J. Phys. Soc. Jpn. 12, 570 (linear response).
- [3] Book chain: Ch.77 (mirror 0/0), Ch.83 (slope -1), Ch.85 (quadratic price), Ch.86 here.